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Autonomous artificial intelligence versus teleophthalmology for diabetic retinopathy

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Abstract

Purpose: To assess the role of artificial intelligence (AI) based automated software for detection of Diabetic Retinopathy (DR) compared with the evaluation of digital retinography by two double masked retina specialists. **Methods:** Two-hundred one patients (mean age 65 ± 13 years) with type 1 diabetes mellitus or type 2 diabetes mellitus were included. All patients were undergoing a retinography and spectral domain optical coherence tomography (SD-OCT, DRI 3D OCT-2000, Topcon) of the macula. The retinal photographs were graded using two validated AI DR screening software (Eye Art ™ and IDx-DR) designed to identify more than mild DR. **Results:** Retinal images of 201 patients were graded. DR (more than mild DR) was detected by the ophthalmologists in 38 (18.9%) patients and by the AI-algorithms in 36 patients (with 30 eyes diagnosed by both algorithms). Ungradable patients by the AI software were 13 (6.5%) and 16 (8%) for the Eye Art and IDx-DR, respectively. Both AI software strategies showed a high sensitivity and specificity for detecting any more than mild DR without showing any statistically significant difference between them. **Conclusions:** The comparison between the diagnosis provided by artificial intelligence based automated software and the reference clinical diagnosis showed that they can work at a level of sensitivity that is similar to that achieved by experts.

Keywords
Diabetes, diabetic retinopathy, macular, artificial intelligence, telemedicine

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Introduction

Over the last decades the prevalence of diabetes mellitus (DM) has significantly increased all over the world. It has been estimated that approximately 415 million people are currently living with diabetes, although it is expected that it will entail an even more serious problem in the future, with more than 640 million people affected by DM in 2040.[1,2] Its relevance would, therefore, be related to a significant increase in the number of chronic and acute DM related complications, with the subsequent impairment in the patient's quality of life and cost increased.[1–3]

Diabetic retinopathy (DR) is a common sight threating complication of DM which remains a leading cause of visual loss in working-age populations.[1–3] DR diagnosis is based on its clinical manifestations of retinal vascular damage.

Diabetic macular edema (DME) is the most prevalent cause of visual impairment among patients with DR and its global prevalence ranges between 2.7% and 11%, depending on type and length of DR.[4,5]

The primary method for diagnosing and assessing DR involves direct and indirect ophthalmoscopy. Nevertheless, the significant advances in the fields of telecommunication and medical imaging have opened new avenues to creating efficient screening strategy for DR. Vujosevic et al.[6] confirmed the relevance of digital images over ophthalmoscopic examination in DR screening. However, independently of the

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DM type, all the DM patients require regular and repetitive annual examinations, with the subsequent time and resource consumption. Although DR fundus photograph screening teleophthalmology programs are mainly performed using mydriatic or non-mydriatic by trained eye technicians, retinal images must be reviewed and graded by well-trained optometrist or ophthalmologist, whose numbers are very scarce compared to the load of patients requiring screening.[7]

AI may be a valuable tool to assist clinical ophthalmologists in the diagnosis and treatment of DR, one of the obvious advantages of AI is its high diagnostic accuracy. Current evidence shows that the accuracy of DR diagnosis can reach up to 90% by using a deep learning mode,[8–10] and 80% or above with a machine learning model.[11–13]

However, AI might be easily affected by the patient source, number and representativeness of sample, algorithm of the AI model, quality of images, use of cameras, and type of algorithm.[14] Moreover, clinical performance of the different AI algorithms varies significantly among different studies.[15,16]

Because data about the use of teleophthalmology and AI for DR screening in Italy are very limited, we wanted to evaluate two different AI algorithms in a real-world scenario and to compare their results versus (vs) two ophthalmologists' specialist in retina.

The current study aimed to compare the performance of two AI algorithms, namely the IDx-DR and the EyeArt, as compared to retinal specialists for detecting DR in patients with DM in a clinical setting.

Methods

Study design

A cross-sectional study conducted on patients recruited or referred to the diabetology clinic at University of Genoa, Italy, from May to September 2019, who were screened for DR using a teleophthalmology program.

The study protocol was approved by the Ethics committee of the University of Genoa, Italy. The study was conducted in accordance with the rules of the Declaration of Helsinki and all the study participants provided written informed consent before starting the study.

Patients

Eligible patients were men or women aged > 18 years, with a confirmed clinical diagnosed of DM, either type 1 or 2, who underwent a teleophthalmology screening program.

Acquisition protocol

Color fundus photos were acquired and transmitted to the Reading Centre of the Ophthalmological Clinic.

Combined color fundus photo and retina scan were obtained with a spectral-domain optical coherence tomography device (SD-OCT, DRI 3D OCT-2000, Topcon). Each patient was seated in a darkened room and both eyes were photographed by a well-trained operator using a two-field protocol: the first one centered on the fovea and the second one on the optic disk (Figure 1). In those patients in whom it was not possible to obtain high quality images without mydriasis a drop of tropicamide 1% was administered 15 min before the examination.

As part of the screening protocol for DR, OCT scans centered on the fovea were acquired at the same time as the color fundus photos, although they were not analyzed in the current study.

Human grading and screening protocol

At the reading center, retinal images were graded by two retinal specialists (DM and MN). In order to minimize errors, ground truth was determined by the grading of two independent graders; in case of discordance images were reexamined until a consensus was obtained. DR and DME were graded according to the international clinical diabetic retinopathy scale,[17] which classifies diabetic

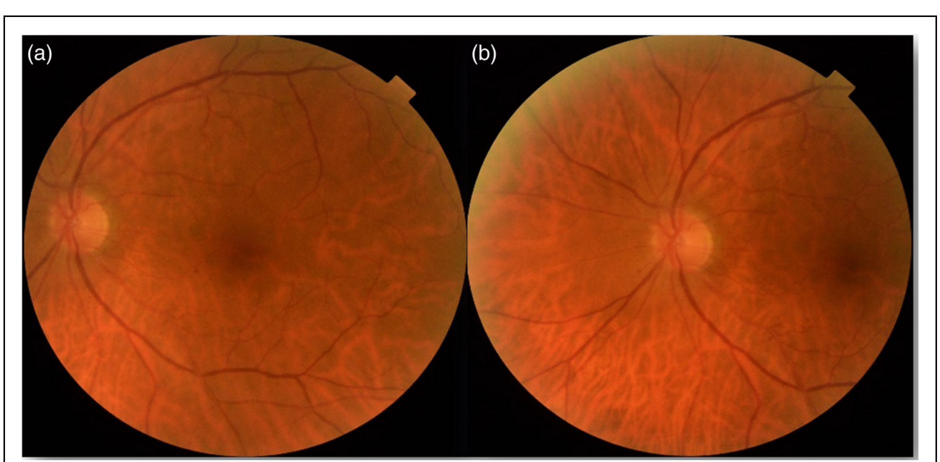

Figure 1. Color fundus retinography. (a). Image centered on the fovea. (b). Image centered on the optic disk.

retinopathy according to the severity of microaneurysms and hemorrhages in 5 steps: no diabetic retinopathy (no abnormalities), mild non-proliferative diabetic retinopathy (NPDR), moderate NPDR, severe NPDR, and proliferative diabetic retinopathy (PDR). Patients without DR or with mild NPDR are recommended to perform a reevaluation within 12 months at the screening service, while patients with moderate to severe NPDR or PDR or with DME were referred to the ophthalmology retina unit at the hospital for a complete ophthalmic examination including OCT and fluorescein angiography, if required. When the quality of the images was "inadequate" for the clinical evaluation (i.e., cataract or corneal opacity) the ophthalmologic evaluation was performed. A report with the results of the screening and the diagnosis with correct follow-up timetable or the need to perform additional studies was sent to patient.

Automated grading

These systems do not require any human intervention in the validation process and the output from the AI systems only takes a few seconds. Because these programs use a cloud-based software, they need an internet connection that provides high speed. In addition, the training datasets for the AI algorithms used in this study were extremely big.[18,19]

Idx. The IDx software (IDX, LLC, Coralville, Iowa, USA) was able to identify signs of DR after image processing involving the use of AI.[20] The output of the IDx software was designed to correlate with the International Clinical of Diabetic Retinopathy (ICDR) criteria. When the quality of the photograph was too low to rule out DR, insufficient image quality feedback was provided. Otherwise, the disease output was reported in four categories: negative, mild DR; moderate DR, and vision threatening DR.

Eyeart. The EyeArt system v2.0 (Eyenuk, Inc., Los Angeles, CA) is a computerized AI device used for assessing color fundus images for DR screening.[21] The program includes different steps for establishing a multi-level classification. According to the results of the analysis, the EyeArt generates an aggregate DR screening recommendation. A "positive" test result would warrant immediately scheduling an appointment with an eye care specialist and a "negative" test results warrants screening again in 9–12 months.[19,21]

Table 1. Baseline demographic and clinical characteristics.

Variable	N = 201
Age, years	
Mean ± SD	65.0 ± 13.1
Gender, n (%)	
Male	108 (53.7)
Female	93 (46.3)
Diabetes, n (%)	
Type 1	29 (14.4)
Type 2	171 (85.1)
Missing data	1 (0.5)
Diabetes duration, years	
Mean ± SD	15.3 ± 10.6
HbA_{1c}, mmol/mol	
Mean ± SD	7.0 ± 1.2
BMI, kg/m^2	
Mean ± SD	27.0 ± 5.2
Blood pressure, mmHg	
Diastolic	
Mean ± SD	79.7 ± 13.8
Systolic	
Mean ± SD	138.4 ± 20.7
Cholesterol, mg/dL	
Total	
Mean ± SD	168.0 ± 37.6
LDL	
Mean ± SD	91.6 ± 30.7
HDL	
Mean ± SD	49.7 ± 16.5
Triglycerides, mg/dL	
Mean ± SD	144 ± 90.3

SD: Standard deviation; BMI: Body mass index; LDL: Low-density lipoprotein; HDL: High-density lipoprotein.

Statistical analysis

Prior to the study it was determined that at least 157 subjects were required, assuming a DR prevalence of 27.6%, with a margin error of 10%, and with a diagnostic sensitivity ≥87%. Given these assumptions and expecting that 20% of the photos will be qualified as insufficient quality by the two AI software's, at least 187 subjects will be needed for this analysis.

The data set was analyzed by the two retina specialists and classified as "gold standard" and by the two AI software. For the analysis, the ability of the AIs software for detecting signs of DR, regardless the grade and/or the presence of DME, was considered. As diabetic retinopathy generally affects both eyes, we defined a case of DR a patient with at least one eye classified as DR. When the software provided a null response in one or both eyes, as it may happens when the image quality is poor, we considered a null prediction for the case.

Results

Based on the tele-screening inclusion on exclusion criteria, both eyes of 201 diabetic patients were imaged. Table 1 summarize the characteristics of the screened patients. All the acquired images were successfully transmitted to the IDx and EyeNet and an output (null, positive or, negative) was collected in all the cases. In 13 (6.5%) and 16

(8.0%) cases the software gave null results for the EyeArt and the IDx platform, respectively.

Firstly, algorithms performances were calculated excluding the null cases. Then, a worst scenario was implemented considering the null cases as misdiagnosed. Table 2 reported sensibility and specificity for both scenarios. For EyeArt, sensitivity and specificity are 100% (95% CI, 89.4%-100%) and 72.9% (95% CI, 65.2%-79.7%), respectively. In the worst scenario, where non gradable images would be all predicted wrongly, performance drops to sensitivity 86.8% (95% CI, 71.9%-95.6%) and specificity 69.3% (95% CI, 61.6%-76.3%). For IDx, Sensitivity and specificity are 94.3% (95% CI, 80.4%-99.3%) and 66.0% (95% CI, 57.8%-73.5%), respectively. In the worst scenario, where low-quality images would be all predicted wrongly, performance drops to 86.8% sensitivity (95% CI, 71.9%-95.6%) and 60.7% specificity (95% CI, 52.8%-60.2%). Independently of the scenario analyzed, there were no statistically significant differences in the algorithms performance. Figure 2 showed the case allocation considering the ground truth and software outputs. The diagnosis of DR was missed by both software in only two (1.0%) cases, whereas for 35 (17.4%) cases both algorithms misdiagnosed non-DR cases. The inter-device concordance between the software was 0.51 (95% CI, 0.39 to 0.64).

Discussion

Because the prevalence of diabetes is rising, the relevance of diabetes-related eye disease increases.[1–3] This fact means that health services must cope with a growing demand with limited means.

Although the burden of DR is significant, early diagnosis and treatment have been shown to be effective in preventing severe visual impairment associated with DR.[3–5] However, early detection and treatment of DR require an effective screening program.

Despite the benefits of screening programs have been unequivocally established, there are different barriers to screening and compliance with current screening recommendations, even in high-resource settings.[22]

The number of ophthalmologists is clearly insufficient, particularly in developing countries, wherein the disease burden is rapidly increasing.[23] Additionally, both direct and indirect health costs may hamper the regular eye care for many patients.[24] Moreover, accessibility of services, particularly in rural areas, due to transportation difficulties, may significantly reduce the utilization of available eye care services.[25]

Ophthalmology is often at the forefront of technological advances in medicine, due mainly to its unique feature of being an open window to the central nervous system, as well as to perform safely surgical procedures with advanced microsurgical techniques.[26] As an example, it is worth mentioning the OCT, a non-invasive method, which provides ophthalmologists the possibility of visualize the retina, in a similar way to a histological section.

Likewise, tele-ophthalmology is one of the fields leading the way for telemedicine, including integration of novel devices enabled with AI to assist in remote patient evaluation and screening.[8–14,26]

We evaluated the performance of two AI systems, which can interpret retinal images for assessing DR in comparison to ophthalmologist. According to the results of our study, the values of sensitivity and specificity ranged from 100%/72.9%, respectively, for the EyeArt device in the best scenario (sensitivity/specificity have been calculated excluding those cases that were not gradable by the software) to 86.8%/60.7%, respectively, for the IDx software in the worst scenario (non-gradable cases have been all predicted wrongly). Both methods have shown a level of sensitivity and specificity equivalent, with not significant differences between them at any scenario.

Artificial intelligence is a field of algorithm-based applications that can simulate humans' mental processes and intellectual activity and enable machines to solve problems with knowledge. Although AI may optimize the care trajectory of patients with chronic disease, it is not widely used in the medical field.[26,27]

There has been a growing need to create systems able to process the enormous amount of data required to make clinical decision. Machine learning algorithms provide computer tools capable to obtain clinically relevant information from patient medical data, to later suggest a diagnosis and / or a clinical management without the face-to-face intervention of the specialist.[28,29] Due mainly to the progress in digitized data acquisition and computing devices, a new strategy called deep learning has emerged as an attractive machine learning method within AI.[30] Deep learning is based on the human brain process operation and involves model architectures called convolutional neural networks.[31] This method has achieved robust diagnostic performance in detecting major ophthalmology conditions, such as DR.[32,33]

Table 2. Sensitivity and specificity (95% CI).

	EyeArt		IDx	p-value
Scenario 1	Sensitivity (95% IC)	100 (89.4 to 100)	94.3 (80.4 to 99.3)	0.25
	Specificity (95% IC)	72.9 (65.2 to 79.7)	66.0 (57.8 to 73.5)	0.44
Scenario 2	Sensitivity (95% IC)	86.8 (71.9 to 95.6)	86.8 (71.9 to 95.6)	0.066
	Specificity (95% IC)	69.3 (61.6 to 76.3)	60.7 (52.8 to 60.2)	0.066

Scenario 1: Specificity and sensibility have been calculated excluding the cases non-gradable by the software. Scenario 2: Non-gradable cases have been considered to be all predicted wrongly.

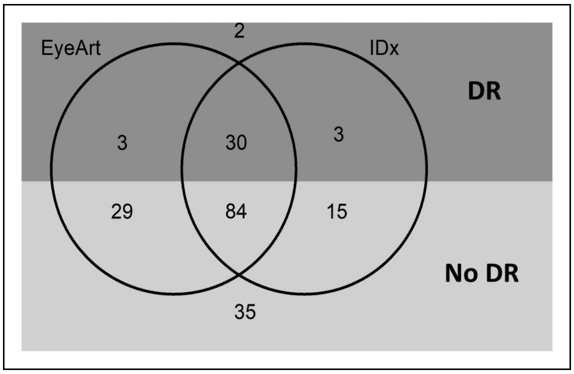

Figure 2. Venn diagram showing agreement between the Human grading (ground truth), the EyeArt, and the IDx systems of detecting and classifying diabetic retinopathy. Dark area: DR ground truth. Light area: no DR ground truth.An overview of the diagnosis determined by human graders in those cases where the software provided null responses is shown in the table.

	Human graders	
Software	DR	No DR
EyeArt*, n (%)	5 (38.5)	8 (61.5)
IDx**, n (%)	3 (18.8)	13 (81.2)

*Percentages were calculated according to the 13 eyes where EyeArt provided null responses.
**Percentages were calculated according to the 16 eyes where IDx provided null responses.
DR: Diabetic retinopathy.

To date, AI techniques have been applied for screening and diagnosing different ophthalmology diseases, including DR, glaucoma, age-related macular degeneration, as well as for predicting the prognosis of other ophthalmic diseases, such as glaucoma.[34]

One of the most promising areas for AI applications in ophthalmology is in DR screening. In fact, AI performance have shown to be better than screening guideline recommendations (typical sensitivity and specificity > 80%).[14]

The values of sensitivity/specificity for detecting DR obtained in our study [Best scenario: 94.3% sensitivity and 66.0% specificity. Worst scenario: 86.8% sensitivity and 60.7% specificity] with the IDx software are slightly lower than those reported by Abràmoff et al [96.8% sensitivity and 87.0% specificity].[20]

Similarly, the results observed in our study with the EyeArt system were slightly lower than those reported by Rajalakshmi et al.,[35] particularly in the worst scenario.

Differences in study protocols may justify these findings.

Finally, COVID-19 pandemic has accelerated AI-based systems as a key asset in streamlining the screening, staging, and treatment planning of sight-threatening eye conditions. The lockdown orders across the world have brought a sudden interest for teleophthalmology and AI devices.[36,37] Moreover, COVID-19 pandemic has accelerated legislative acceptance of telehealth at least temporarily, and has helped further its availability and usage.[37]

Several limitations should be taken into consideration when interpreting these results. The main one is the fact that the "Gold standard" method was as subjective as "Expert opinion". Nevertheless, retinal images were graded by two experienced and well-trained retinal specialists and in case of discordance images were reexamined until a consensus was obtained. On the other hand, a characteristic and possible limitation of AI software is that researchers and users typically know the inputs and outputs, but even those who design the algorithms, cannot understand how variables are being combined to make decisions. Additionally, this was a cross-sectional study and, therefore, progression cannot be assessed. Furthermore, this was a single-center study that included a limited number of patients, which was conducted on a clinical setting. In addition, two specific algorithms were assessed, so special care must be taken when generalizing the results.

Finally, the agreement on edema detection was not investigated.

In conclusion, the results of study provided evidence about the accuracy and applicability of AI for diagnosis DR in a clinical setting. Additionally, the current study also observed that the sensitivity/specificity values obtained with both AI devices were like that achieved by experts.

Although AI devices may be an effective solution for detecting DR and improving the access of people to specialized ophthalmology attention, how the patient perceives the fact of being treated by a machine instead of a specialist should be considered.

Moreover, there is no doubt that AI is a useful support for screening of several ophthalmic diseases, but in no case can it replace the role of ophthalmologists in clinical diagnosis and management.

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